

CAN Bus Assignment Part 1

Technical Report

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Introduction

This assignment focuses on the CAN bus protocol. For part 2, two Arduino boards are communicating with each other via the CAN bus, using the mcp2515 modules and their library functions. We must send a command from a **transmitter** Arduino to a **receiver** Arduino and control LEDs using a joystick.

Note: Some liberties were taken when creating this assignment. Instead of three boards, two are used with two LEDs attached to the receiving end.

Hardware

- 2 Arduino Uno R3 based Sparkfun Redboards
- 2 mcp2515 CAN modules
- 2 LEDs

Software

The software uses the Arduino abstraction layer for handling embedded functions. Using the mcp2515 library to interface with the can modules. Also the Sparkfun Qwiic Joystick library is used to read joystick inputs

There are two Arduinos at play, the “transmitter” and the “receiver”. The transmitter, at address 0x43, sends user commands via the CAN bus to the receiver. The user moves the joystick to a position and the transmitter sends a command depending on the joystick’s place. The receiver interprets the command and reacts accordingly. Either blinking a LED, turning another on etc.

Also, as to not flood the network unnecessarily. I implemented a solution which sends a command only once per position change. Therefore we are not constantly spamming the network with the same message

Testing and measurements

I can confirm visually that my programs function correctly. Both LEDs can work simultaneously and program does not get stuck, even though I am not using FreeRTOS here, which could improve performance and eliminate the polling aspect of the program.

Conclusion

The program is functioning as intended. I think I did very well on solving the network flooding problem. But I could improve on code readability and maintainability by separating my program in modules. Also FreeRTOS could help improve performance. I should look into that.